ACADEMIC & AGRICULTURAL PORTFOLIO DOCUMENTATION

Analyzing Rainfall Impact on Agricultural Productivity

DOMAIN	PRIMARY TOOL	CERTIFICATION	PROJECT TYPE
Data Analytics & BI	Power BI Desktop	Hero Vired	Capstone Project

1. PROJECT OVERVIEW & EXECUTIVE SUMMARY

This project analyzes multi-decade weather patterns and state-level agricultural outputs to evaluate how precipitation variance influences crop yields. Developed using Power BI Desktop as part of the Hero Vired Data Analytics curriculum, the interactive dashboard tracks monthly monsoon rainfall and correlates it directly with crop productivity metrics across Indian states from 1966 onwards.

2. DASHBOARD METRICS & PROBLEM STATEMENT

Unpredictable rainfall patterns cause severe variance in crop output across different regional belts. To quantify this impact, key monthly precipitation indicators were established alongside crop yield distributions across states:

Monsoon Month	Avg Rainfall (mm)	Analyzed Crop Types	Interactive Slicers
June	191.55 mm	Barley Yield (Avg by State)	State Code (1 to 7+)
July	304.36 mm	Cotton Yield (Sum by State)	Historical Years (1966–1972+)
August	268.22 mm	Castor Yield (Sum in Kg/ha)	Geographic Regional Filtering
September	173.56 mm	Multi-crop Comparison	Dynamic Slicing & Drilldown

3. TECHNICAL IMPLEMENTATION & VISUALIZATIONS

- **KPI Scorecards:** Built top-level card visualizations displaying peak monsoon rainfall in July (304.36 mm) down to late-season precipitation in September (173.56 mm).
- **Barley Yield Bar Chart:** Plotted average barley yield across states, highlighting top-producing states such as Rajasthan, Haryana, Uttar Pradesh, and Punjab.
- **Cotton Yield Area Chart:** Visualized cumulative cotton production trends across key agricultural belts including Tamil Nadu, Gujarat, Haryana, and Andhra Pradesh.
- **Castor Yield Horizontal Chart:** Detailed castor productivity distribution across lead producing states including Gujarat, Karnataka, Assam, and Rajasthan.

4. KEY INSIGHTS & ANALYTICAL FINDINGS

- **Monsoon Peak Alignment:** Precipitation peaks sharply during July (304.36 mm) and August (268.22 mm), identifying these two months as the critical moisture window for Kharif crop germination and growth.
- **State-Level Productivity Concentrations:** Northern states (Rajasthan, Haryana, UP) dominate average Barley production, whereas Western and Southern states (Gujarat, Tamil Nadu, Karnataka) show higher cumulative yields for Cotton and Castor.
- **Historical Longitudinal Scope:** Filtering across historical year slicers (1966 onwards) enables long-term trend analysis of agricultural resiliency against fluctuating precipitation cycles.

> **Deliverables Included:** Power BI Project File (`.pbix`), Processed Dataset, GitHub Codebase Repository, and PDF Executive Report for University Admissions.

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Average Rainfall in JUNE	Average Rainfall in JULY	Average Rainfall in AUG	Average Rainfall in SEP
191.55	304.36	268.22	173.56









